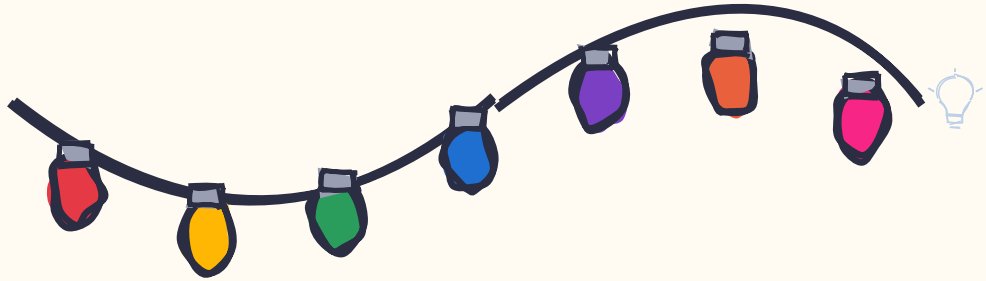
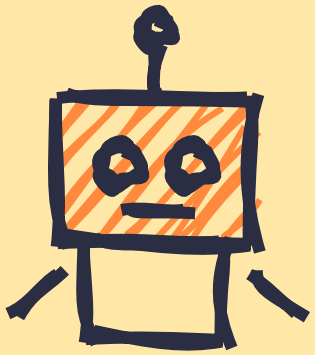
Junior Makers · Week 1

Series & Parallel Circuit Lab

How does electricity travel, and which path does it take?

Ages 8-11 **Breadboard + multimeter** **PhET simulator**





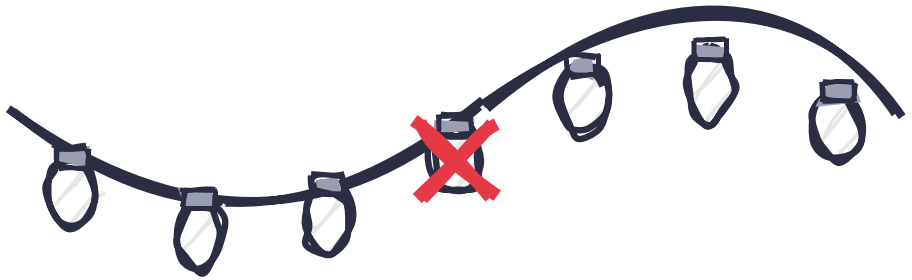


Did you know?
Electricity in a wire moves the signal at nearly the speed of light!

Mystery time!

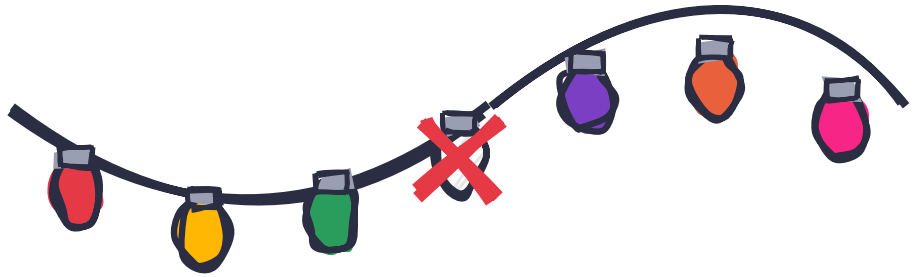
The case of the broken fairy lights

String A: one bulb breaks...



...and **ALL** the lights go out!

String B: one bulb breaks...



...and the others **keep shining!**

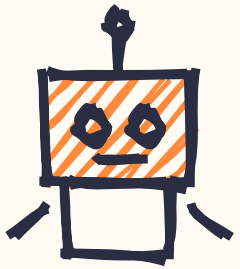
Why? The bulbs look the same. What is different?

In real life

Think of the lights on houses at Eid or at a wedding. Sometimes a whole string goes dark!

Our learning aims

By the end of today, I can…



1. Explain what electricity is and what every circuit needs
2. Build a **series** circuit and a **parallel** circuit
3. Measure voltage with a **multimeter** and compare results
4. Explain which circuit our **homes** use, and why

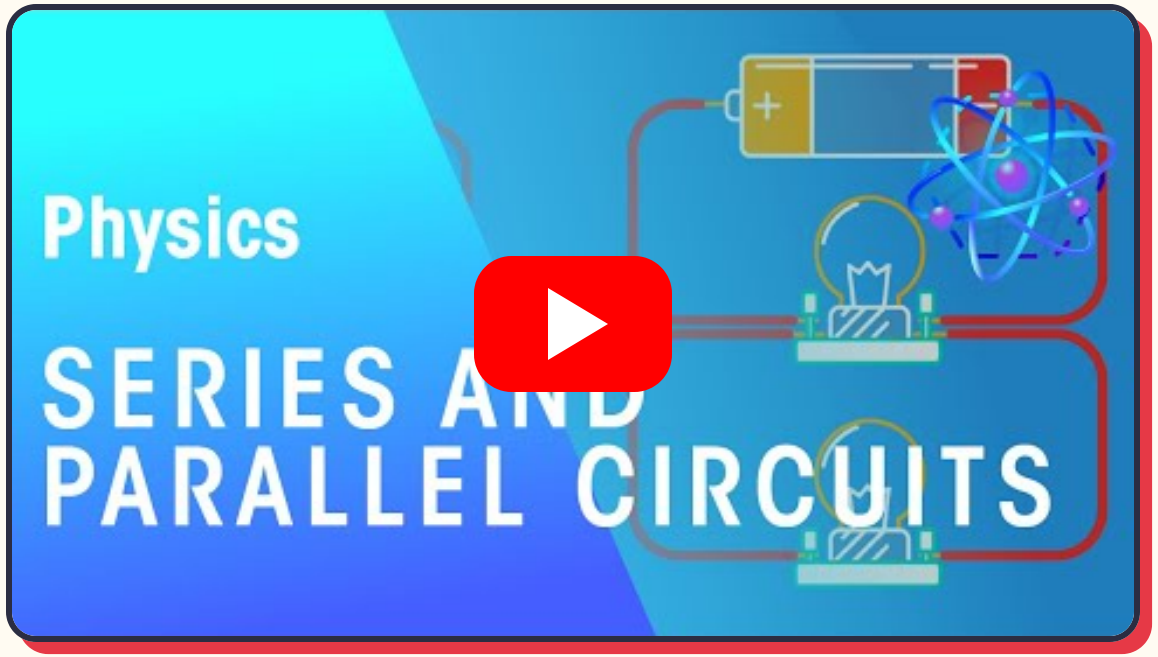
Keywords

circuit · current · voltage · series · parallel · conductor · insulator

LEARN

Watch and think

Video time!



Series and Parallel Circuits

FuseSchool · about 5 minutes · youtu.be/8Z0jhQeYDUE

Watch for...

1. In a **series** circuit, how many paths can the electricity take?
2. What happens to the other bulbs when one breaks in a series circuit?
3. Why are houses wired in **parallel**?

Scan to watch

After the video: Draw a tiny series circuit and a tiny parallel circuit on your hand or a sticky note. Swap and check.

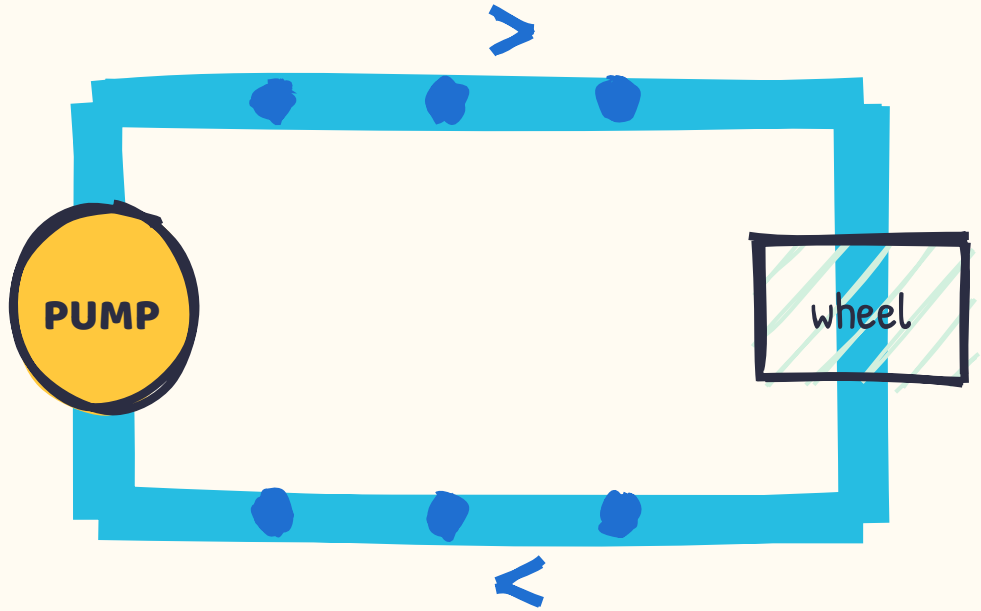
LEARN

Let's start at the beginning

What is electricity?

Everything is made of tiny particles. Some of them, called **electrons**, can move through metal wires.

When lots of electrons flow together in one direction, we call it an **electric current**. The current carries energy to light bulbs, motors and screens.



pump = battery **pipe** = wire **water** = current

Think of it like water

A battery is like a pump. Wires are like pipes. The current is like water flowing round and round.

Energy on the move

Where does electricity come from?



Batteries

Chemical energy inside turns into electricity. Remotes, toys, torches.



Solar panels

Sunlight turns into electricity. Rooftops and calculators.



Power stations

Big generators send electricity to every wall socket.



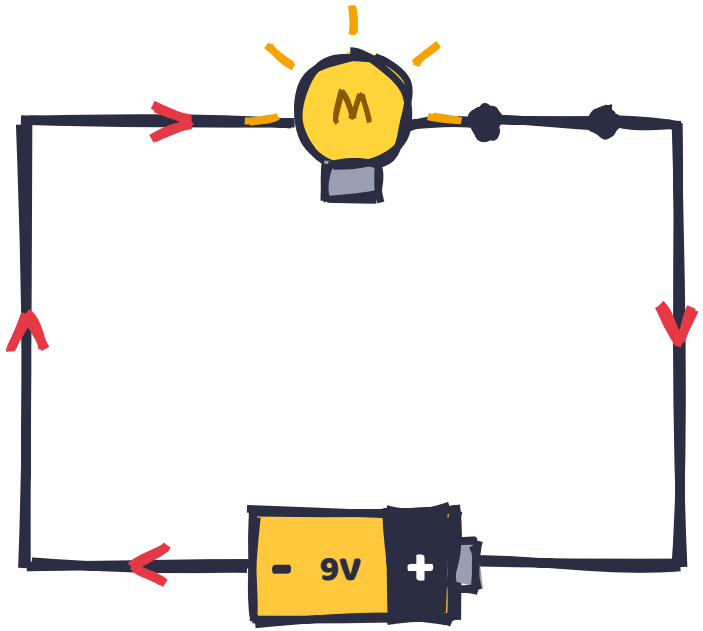
Safety first: wall sockets can seriously hurt you. In our lab we only use small, safe **9V batteries**.

LEARN

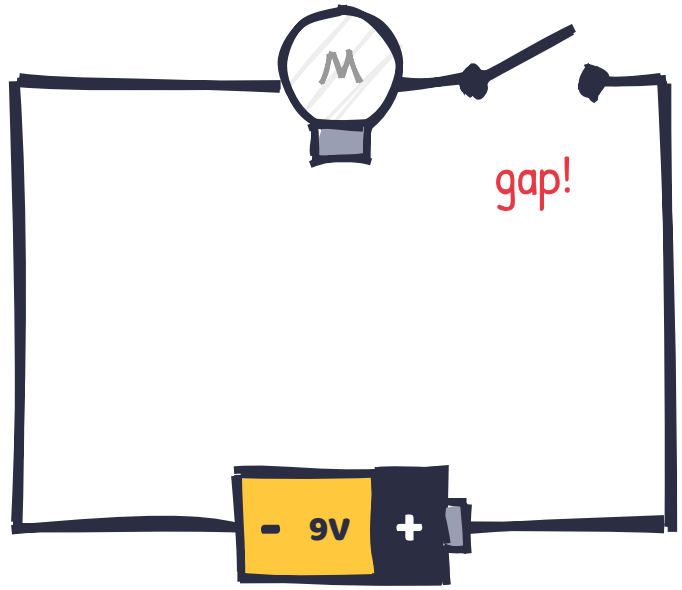
The magic loop

What is a circuit?

Closed circuit: it works!



Open circuit: no light



In real life
A light switch opens and closes a gap in the loop!

A **circuit** is a complete loop. Current leaves the battery, goes through the bulb and comes back. Any **gap** stops it.

Engineer's language

Parts of a circuit and their symbols



Power source

battery or cell



Conductor

wires carry the current



Load

bulb, LED, motor, buzzer



Switch

opens or closes the loop



LED

a light that only works one way

Engineers draw circuits with **symbols** instead of pictures. It is faster, and engineers all over the world understand it, like a secret code!

Try it!

Draw today's circuit using only symbols in your notebook.

Let electricity through, or not?

Conductors and insulators

Conductors: let current flow

Copper wire · Metal spoon · Aluminium foil · Coins · Paper clip

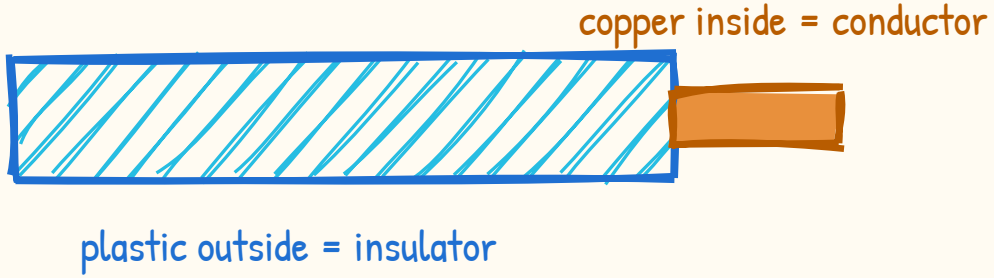
Mostly metals

Insulators: block current

Plastic ruler · Rubber eraser · Wooden stick · Paper · Glass

Plastic, rubber, wood

Look inside a real wire:

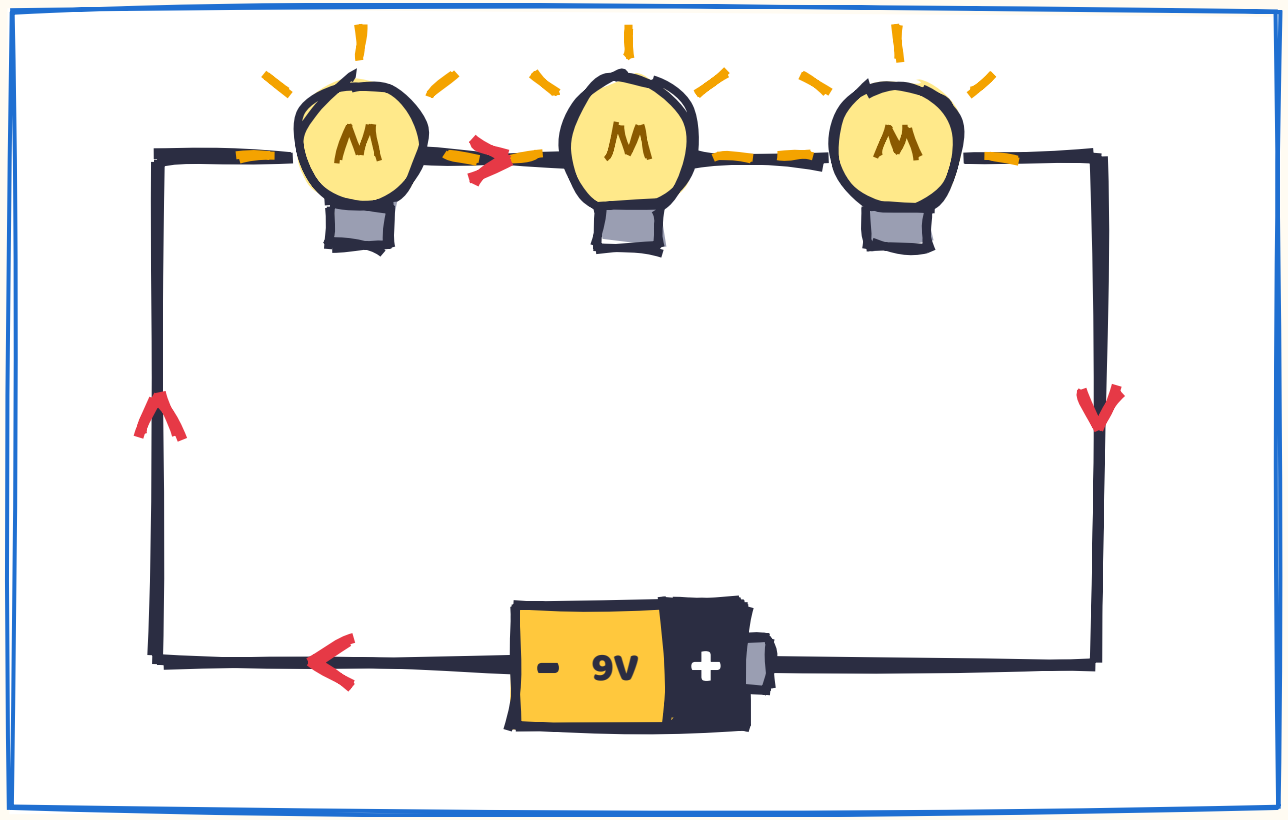


Quick test

Put a spoon, then a ruler, in the gap of a circuit. Which one lights the LED?

One single path

Series circuit



Series = components one after another, in a line.

One path for the current, like a single-lane road.

One breaks, all stop. The loop is broken.

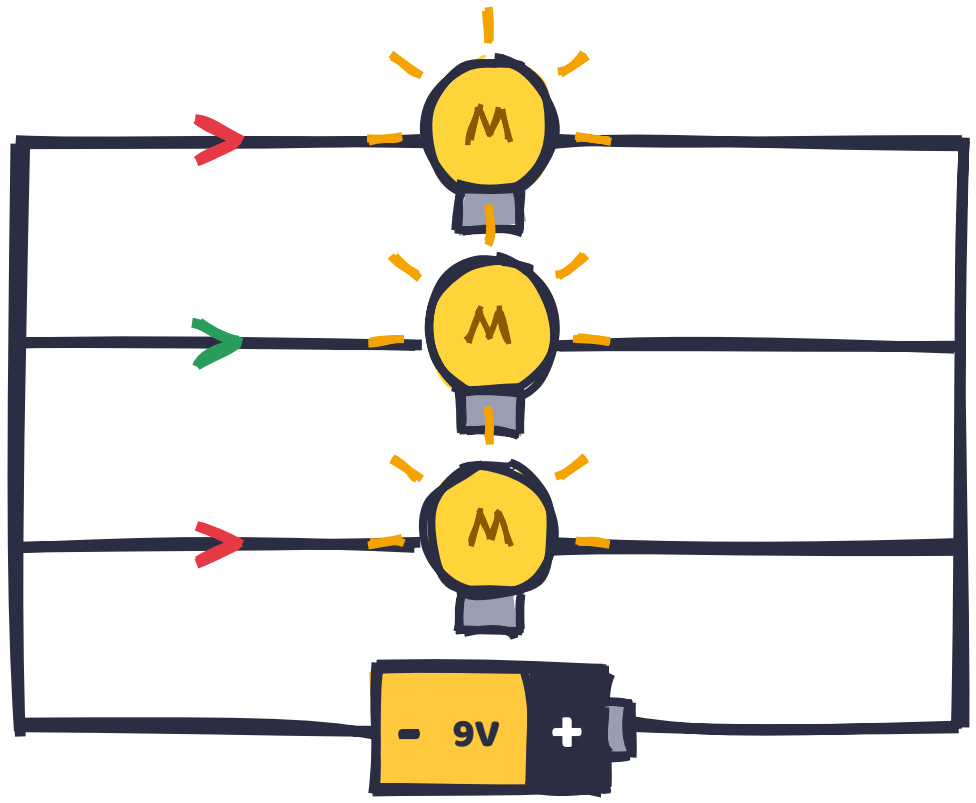
Bulbs share the battery's push (voltage), so each glows **dimmer**.

In real life

Batteries in a TV remote or torch sit in series to give more push. Old fairy lights were wired in series too!

Many paths

Parallel circuit



Many paths: each bulb has its own lane back to the battery.

One breaks, others keep working.

Each bulb gets the full push (voltage), so they stay **bright**.

In real life

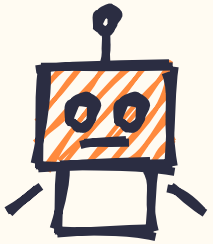
Your home is wired in parallel. That is why the fan keeps running when a bulb in the same room fuses!

LEARN

Side by side

Series vs parallel

	Series	Parallel
Paths for current	One path	Many paths
If one part breaks	Everything stops	The rest keep working
Brightness	Dimmer (share voltage)	Bright (full voltage each)
Where we see it	Torch batteries, old fairy lights	Homes, cars, classrooms



In real life
Car headlights are in parallel: if one fails, the other still lights the road.



Series is a single-lane road: one blockage stops everyone. **Parallel** is a road with many lanes: traffic finds another way.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

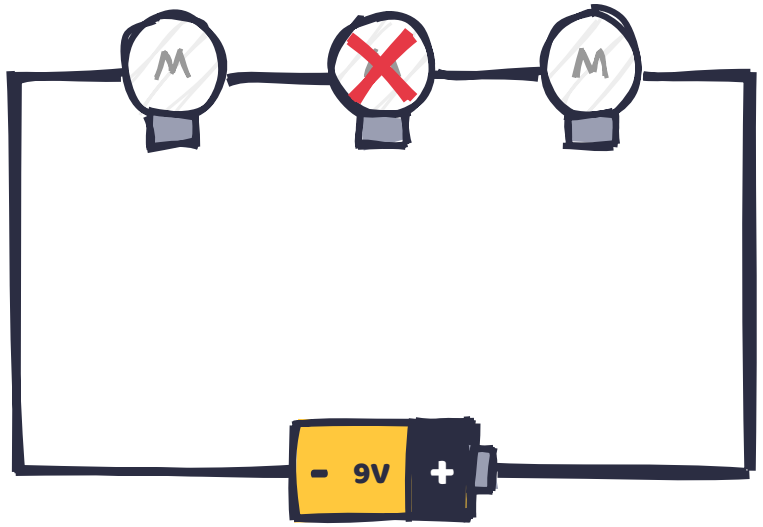
1. What happens when one bulb breaks?
2. Measuring voltage with a multimeter
3. Lab missions

MAKE

Predict, then test

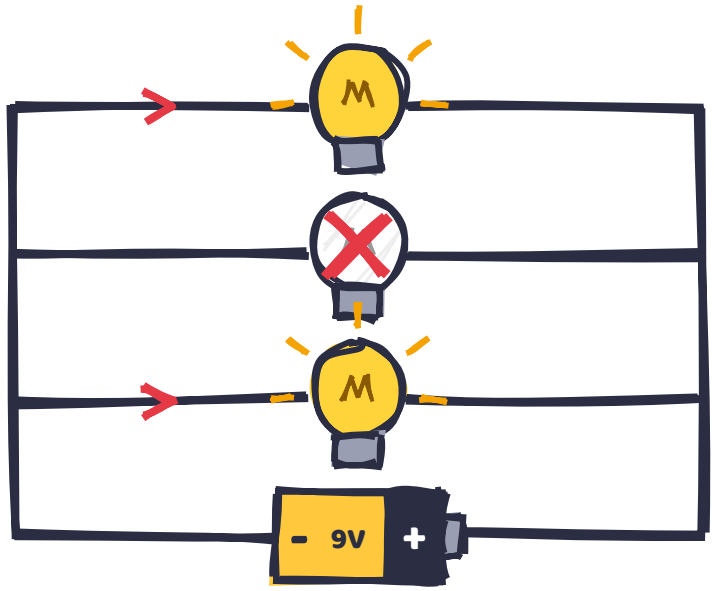
What happens when one bulb breaks?

Series: one breaks...



...the whole loop is broken. All OFF.

Parallel: one breaks...

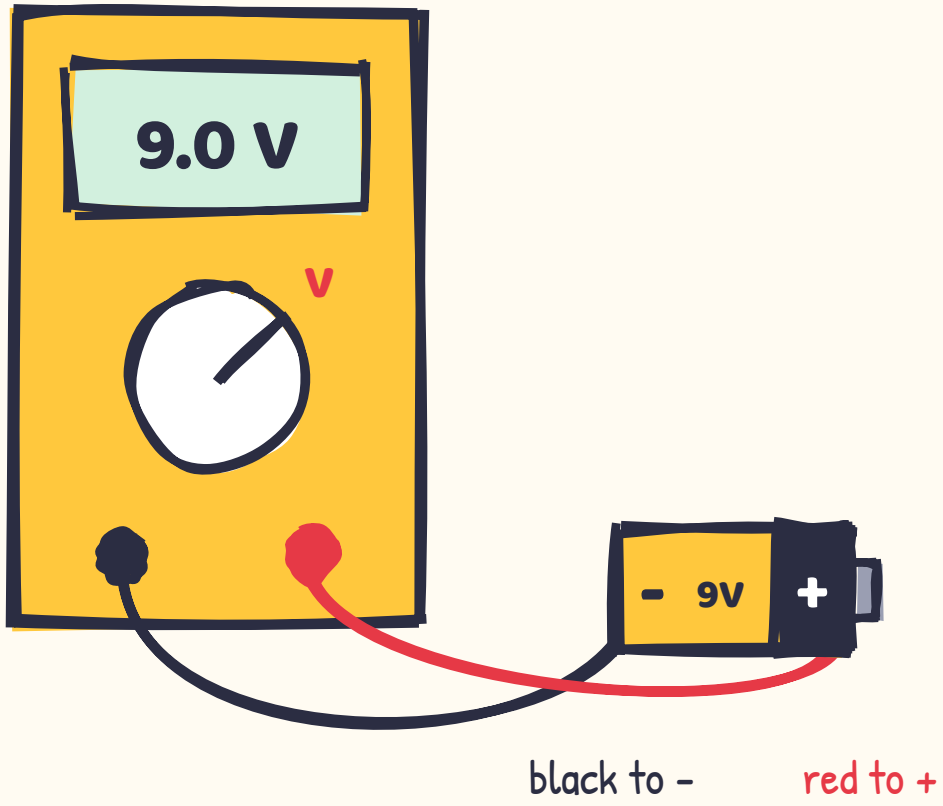


...other paths still work. Others ON.

Now solve the mystery: which fairy light string was series, and which was parallel?

Scientist's tool

Measuring voltage with a multimeter



1. Turn the dial to **V** (DC volts, 20V range).
2. **Black** probe to **-**, **red** probe to **+**.
3. Read the number: that is the **voltage**, the battery's push.
4. Measure across each LED in series, then in parallel.

Safety rule
Only measure batteries in the lab. Never put probes in a wall socket.

In real life
Electricians check every house wire with a meter like this.

MAKE

Your turn, engineers!

Lab missions

Mission 1: Simulate

Build both circuits in the PhET simulator. Break a bulb in each and watch.

Mission 2: Series

On the breadboard: 9V, resistor and 2 LEDs in one line. Measure the voltage on each LED.

Mission 3: Parallel

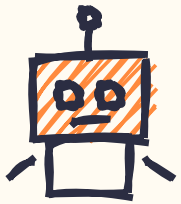
Give each LED its own path (and its own resistor). Measure again.

Mission 4: Record

Fill the table: voltage, brightness, what happens when one LED is removed.

Teacher tip

Use red LEDs with a 220 ohm resistor. In series, use 2 LEDs so the 9V battery can light both.



Circuit	LED 1 (V)	LED 2 (V)	Brightness	Remove 1 LED?
Series	____	____	____	____
Parallel	____	____	____	____

Show what you know!

Recap and challenge

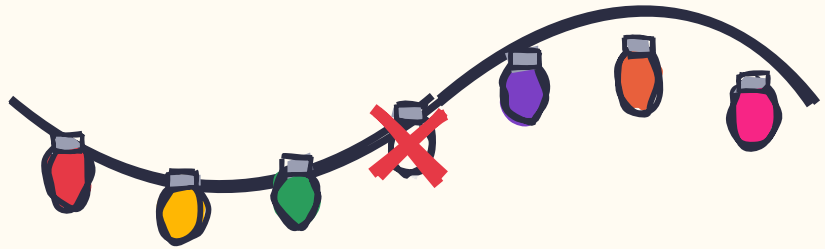
Exit ticket

- [x] Name 3 parts every circuit needs.
- [x] In which circuit do all lights go out if one breaks?
- [x] Why are homes wired in parallel?

Design challenge

Design the lights for a 3-room house. Draw it with symbols. Which circuit will you use, and where will the switches go?

Mystery solved!
String A was SERIES (one path). String B was PARALLEL (many paths).



Next week: a two-player **Reaction Timer** game with a scoreboard!